GRP Research Memorandum

VA + PA Queue Washout Cross-Jurisdiction Analysis

Does the AEP Ohio pattern repeat in Dominion and PPL/Constellation territory?

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Baseline: What the AEP Ohio Filing Said

In a PJM system impact study update, American Electric Power (AEP) disclosed that its Columbus data center interconnection cluster experienced a significant queue washout following the implementation of a take-or-pay tariff:

Metric	Value
MW withdrawn from uncommitted queue	24,300 MW
Contracted load before washout	30,000 MW
Bankable contracted load after washout	5,700 MW
Avg. COD extension across affected positions	+14 months
Source	AEP PJM System Impact Study Update (2025–2026)

Virginia — Dominion Energy (ticker: D)

Finding: Different regulatory layer — no comparable washout signal.

The VA SCC case PUR-2022-00073, titled *In the Matter Considering Utility Distributed Energy Resource Interconnection*, is about **distributed energy resource (DER)** interconnection reform — rooftop solar, small battery storage, and co-op interconnections. It is not the same regulatory instrument that would capture utility-scale data center queue activity.

Dominion's most recent cluster pilot report (October 2025, filed with VA SCC) disclosed:

> *“As the Pilot is now closed with no projects enrolled, the Company, in accordance with the Order, submits its fifth and final report.”*

Dominion's **large-scale transmission interconnection queue** (where hyperscale data centers would sit) is managed through PJM's RTEP process, not through VA SCC. The VA SCC regulates Dominion's distribution system, not PJM-level transmission. No take-or-pay tariff equivalent to AEP's is visible in VA SCC filings.

VERDICT	No comparable queue washout found. Different regulatory layer from AEP Ohio. To assess Dominion's true data center queue exposure, PJM zone-level queue data (DOM zone, RTEP filings) would be required — not available via VA SCC.

Pennsylvania — PPL Electric (ticker: PPL) & FirstEnergy PA

Finding: Counter-signal — PPL is actively welcoming 9 GW of data center pipeline.

The PA PUC held a formal en banc hearing on *Interconnection and Tariffs for Large Load Customers* on April 24, 2025 (Docket No. M-2025-3054271). Unlike Virginia, Pennsylvania regulators convened this hearing specifically because the data center interconnection wave is large enough to require a new tariff framework.

PPL Electric VP Joseph Lookup testified:

> *"PPL Electric is directly seeing the growth of data center development in Pennsylvania, with requests in advanced stages in excess of 9 GW of new load as reported during PPL Corporation's year-end earnings call. To put this into perspective, PPL Electric's current summer peak is 7.5 GW, and the new data center requests are poised to more than double PPL Electric's system peak within the next 5–6 years."*

> *"PPL Electric has invested in the reliability and resiliency of its transmission system to better serve its customers. An additional benefit of this investment is that PPL Electric now stands ready to serve this influx of load with large load customers only having to cover the incremental cost of interconnecting their facilities."*

> *"The Company estimates that the first gigawatt of interconnected load will reduce other customers' transmission costs by 10%."*

PPL is explicitly framing data center load as beneficial to all ratepayers. They have capacity headroom from prior transmission investment. **No take-or-pay tariff has been implemented** — the PA Office of Consumer Advocate (OCA) is advocating for one as a future protection, which means the washout risk is prospective rather than realized:

> *"The objectives of transparency, non-discriminatory access, fair cost allocation, and protection from stranded assets are achievable through a dedicated large load tariff."*

FirstEnergy PA (covering 2.1 million customers across 56 of 67 PA counties) also testified that it is adapting its load study process for data center characteristics but described the situation as an evolving challenge, not a washout event.

VERDICT	No queue washout. PPL has 9 GW of active pipeline, capacity headroom, and a favorable regulatory posture. The washout risk the OCA is flagging is a future policy risk (take-or-pay implementation), not a current withdrawal event.

Cross-Jurisdiction Comparison

Jurisdiction	Utility	Queue Status	Take-or-Pay	Washout Risk
Ohio	AEP (AEP)	Washout confirmed (24.3 GW withdrawn)	Yes — triggered washout	**Already realized**
Virginia	Dominion (D)	DER queue only; PJM manages utility-scale	No equivalent visible	Low (different layer)
Pennsylvania	PPL (PPL)	9 GW active pipeline; capacity headroom	No — OCA pushing for it	Future risk; not current
Pennsylvania	FirstEnergy PA	Active; adapting processes	No — considering it	Low-medium

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Synthesis: Does This Break the Sector Thesis?

No — and the cross-check clarifies something important.

The AEP Ohio washout appears to be **AEP-specific**, driven by two compounding factors:

1. AEP's specific take-or-pay tariff design for the Columbus data center cluster
2. That specific cluster having unusually speculative queue composition (30 GW filed vs. only 5.7 GW bankable demand)

What this means for ticker-level differentiation:

Ticker	Signal	Rationale
AEP	**NEGATIVE**	24.3 GW washout confirmed. EPS guidance at risk from reduced load growth assumptions in Columbus cluster zone.
PPL	**POSITIVE**	9 GW pipeline, capacity headroom from prior capex. Expects to benefit from rate dilution for existing customers.
D	**NEUTRAL / MONITOR**	No VA SCC signal. True exposure requires PJM DOM-zone queue data (RTEP filings) — not captured here.
CEG	**NEUTRAL**	Constellation operates nuclear plants, not distribution. PA regulatory exposure is indirect via FirstEnergy territory.

Forward-Looking Risk to Monitor

If the PA PUC implements a PPL take-or-pay tariff modeled on AEP's design (which the OCA is actively advocating), PPL's 9 GW pipeline could see a similar washout event. That would be a **negative inflection for PPL** — but it requires a specific PUC ruling to trigger and is likely 12–18 months away from becoming a realized event, not an immediate earnings risk.

The Harvard Electricity Law Initiative paper cited by PA PUC Vice Chair Barrow (*"Extracting Profits from the Public: How Utility Ratepayers are Paying for Big Tech's Power"*) suggests the political pressure for stronger cost-allocation protections is building. This is a regulatory trend to track across all PJM-footprint utilities.

Primary Sources

• AEP: PJM System Impact Study Update (Columbus data center cluster, 2025–2026)

• VA: VA SCC Docket PUR-2022-00073 — In the Matter Considering Utility DER Interconnection. Dominion Energy quarterly cluster pilot reports (Jan 2024 – Oct 2025). Retrieved via VA SCC DocketSearch REST API.

• PA: PA PUC Docket M-2025-3054271 — En Banc Hearing on Interconnection and Tariffs for Large Load Customers, April 24, 2025. Testimonies of PPL Electric (Joseph Lookup, VP), FirstEnergy PA (Kelly Gower, VP), Office of Consumer Advocate (Darryl Lawrence, Acting CA), PECO Energy, Amazon Data Services, Data Center Coalition.

• All documents retrieved programmatically via Grid Realization Pipeline (GRP) on 2026-07-12.